

Question				Marking details	Marks Available					
					AO1	AO2	AO3	Total	Maths	Prac
8	(a)	(i)		Skin (connective tissue) / skin flora (bacteria)/ blood clotting / inflammation/ phagocytes / macrophage lysozyme/ stomach acid/ low pH in stomach Any 3 for 1 mark	1			1		
		(ii)		{Many organisms/ named organism} live {in/on} the (human) body	1			1		
		(iii)		Cell lysis / cell bursts	1			1		
		(iv)		Any three × (1) from: • {Macrophages/ phagocytes} {engulf virus infected cell / carry out phagocytosis}/ antigen presentation (1) • clonal expansion/ T cells {proliferate/divide many times} / (1) • Cytokines stimulate phagocytosis (1) • Killer T cells cause cell lysis (1)	1	2		3		
	(b)	(i)		- Antibiotics can act on {cell wall of bacteria/ or description of/ hydrolyse peptidoglycan} (1) - Antibiotics affect {metabolism/protein synthesis} by bacterial cell (1) - Viruses don't have a {cell wall / metabolism} so not affected by antibiotics (1)	2		1	3		
		(ii)		7250 (mg) (2) If incorrect award 1 mark for any combination of three parts of equation $25\text{mg} \times 14.5 \times 2 \times 10$		2		2	2	